

Pima Indians Diabetes prediction

A Logistic Regression Project

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Project Type: Logistic Regression

Dataset: Pima Indians Diabetes Database

Dataset Source: Kaggle

Kaggle Link: <https://www.kaggle.com/datasets/jamaltariqcheema/pima-indians-diabetes-dataset/data>

Code Notebook (Colab): https://colab.research.google.com/drive/1RR1_VeFcb1xLvUKI0VI5s7_sf6rdSsch?usp=sharing

2. Objective

Build a logistic regression model to predict whether a person has diabetes (Outcome: 0 = No diabetes, 1 = Diabetes) based on the available clinical input features - pregnancies, glucose, blood pressure, skin thickness, insulin, BMI, diabetes pedigree function, and age. The goal is to understand which clinical features most strongly influence the predicted diabetes risk and to evaluate how well the model classifies unseen cases (measured via accuracy, precision, recall, F1-score, and ROC-AUC).

Workflow:

Load the Kaggle CSV > inspect the dataset > perform exploratory analysis > identify and replace invalid zeros > split the data into training and test sets > median imputation > feature scaling > train balanced Logistic Regression > evaluate the model > interpret model coefficients.

3. Dataset Overview

The dataset is the Pima Indians Diabetes Database from Kaggle. It contains 768 rows and 9 columns of clinical measurements, with a binary target column (Outcome) indicating whether the person has diabetes.

Property	Value
Rows	768
Columns	9
Target variable	Outcome (int64; 0 = No diabetes, 1 = Diabetes)
Integer features (int64)	Pregnancies, Glucose, SkinThickness, Age
Float features (float64)	BloodPressure, Insulin, BMI, DiabetesPedigreeFunction
Categorical features	None

Sample records (first 5 rows):

Pregnancies	Glucose	BloodPressure	DiabetesPedigree Function	Age	Outcome
6	148	72.0	0.627	50	1
1	85	66.0	0.351	31	0
8	183	64.0	0.672	32	1
1	89	66.0	0.167	21	0
0	137	40.0	2.288	33	1

Note: SkinThickness, Insulin and BMI columns are hidden in the printed preview of the notebook output.

4. Data Quality and Cleaning

Check	Before Cleaning	After Cleaning
Rows	768	768
Duplicate rows	0	0
Missing values (all columns)	0	0
Invalid zeros in Glucose, BloodPressure, SkinThickness, Insulin, BMI (replaced with NaN)	0	0

Outlier check:

The boxplots of the numeric features (see Section 6) show points beyond the whiskers in several columns, most clearly in Insulin, and also in SkinThickness, BloodPressure, BMI and Age. No rows were removed, so all 768 rows were retained (614 for training and 154 for testing).

Cleaning decisions:

- No duplicate rows were found, so no rows were dropped for duplication.
- No missing values were found in the dataset, and the invalid-zero check on Glucose, BloodPressure, SkinThickness, Insulin and BMI (zeros replaced with NaN) left 0 missing values in each of these columns.
- Median imputation was fitted on the training data and then applied to the test data, so no information from the test set leaks into training.
- StandardScaler was fitted on the imputed training data and applied to both the training and test sets.

5. Descriptive Analysis

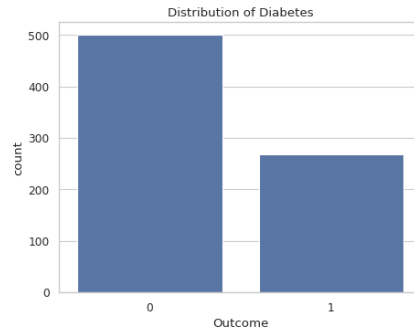
Summary for the target variable (Outcome):

Category	Meaning	% of Data
0	No diabetes	65.10%
1	Diabetes	34.90%

About two thirds of the records belong to the 'No diabetes' class and about one third to the 'Diabetes' class, so the classes are imbalanced. For this reason the Logistic Regression model was trained with `class_weight='balanced'`.

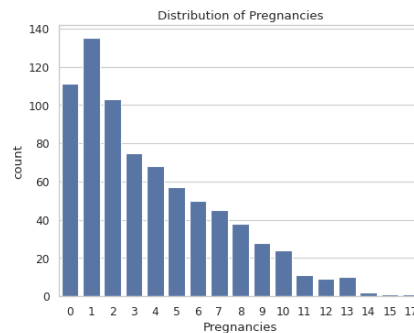
6. Visual Analysis

Chart 1: Distribution of Diabetes (Univariate)



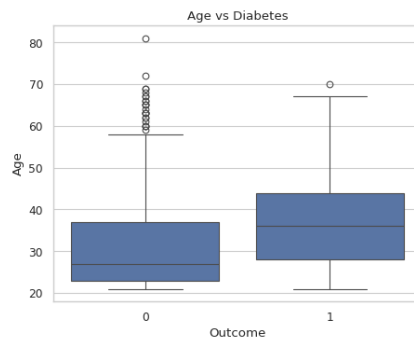
Observation: About 65% of the records have no diabetes (0) and about 35% have diabetes (1), confirming the class imbalance.

Chart 2: Distribution of Pregnancies (Univariate)



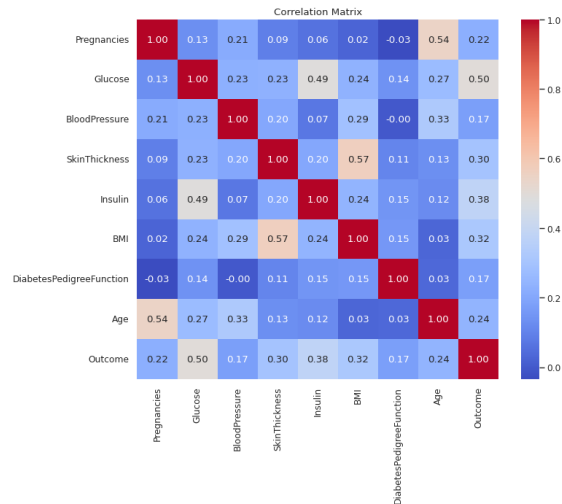
Observation: The distribution is right-skewed - most records have 0 to 3 pregnancies, with a long tail up to 17.

Chart 3: Age vs Diabetes (Bivariate)



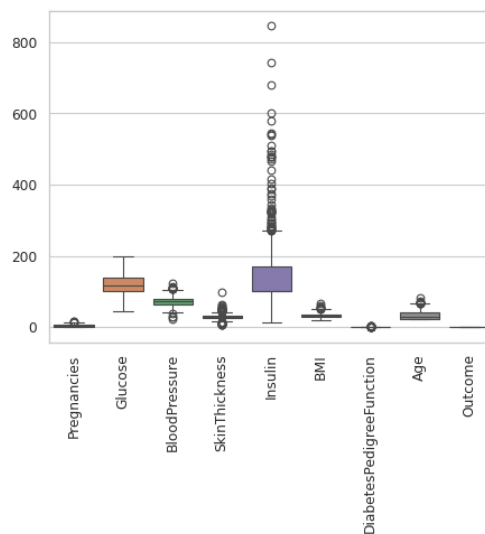
Observation: The diabetes group (1) has a higher median age and a higher middle spread than the non-diabetes group (0).

Chart 4: Correlation Matrix



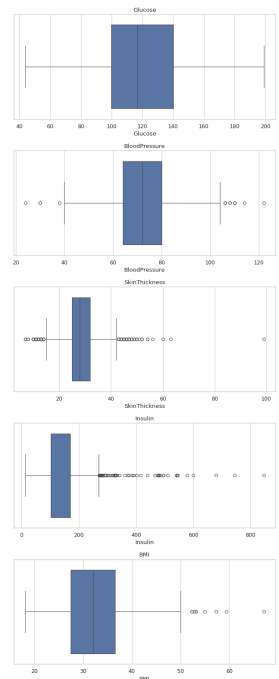
Observation: Glucose has the strongest correlation with Outcome (0.50), followed by Insulin (0.38) and BMI (0.32). Among the input features, SkinThickness-BMI (0.57) and Age-Pregnancies (0.54) are the most strongly related pairs.

Chart 5: Outlier Detection Boxplots



Observation: Insulin shows many high points above the upper whisker; several other features also have a few points beyond their whiskers.

Chart 6: Boxplots After Treating Invalid Zeros



Observation: Glucose, BloodPressure, SkinThickness, Insulin and BMI after the invalid-zero treatment. Insulin and SkinThickness still show high-end points, which were kept in the data.

7. Model Building and Evaluation

The data was split 80:20 into training and test sets with stratification (random_state=42), giving training shapes of (614, 8) and test shapes of (154, 8). Median imputation was fitted on the training data and applied to the test data, StandardScaler was fitted on the imputed training data and applied to both sets, and a Logistic Regression model (class_weight='balanced', max_iter=1000) was trained and evaluated on the held-out test set.

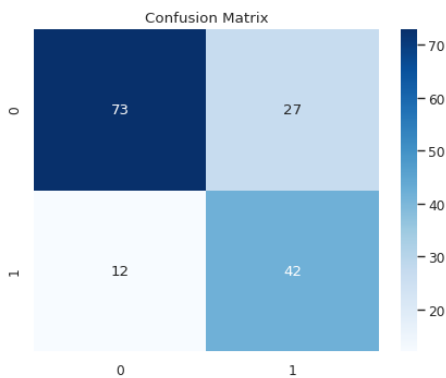
Metric	Value
Accuracy	0.7468
Precision	0.6087
Recall	0.7778
F1-score	0.6829
ROC-AUC	0.8272

The model correctly classifies about 74.7% of the test cases. A recall of 77.8% means it identified a relatively large share of the actual diabetes cases, which is useful for a screening-oriented task, while the precision of 60.9% shows that some non-diabetic cases are flagged as diabetic.

Classification report:

Class	Precision	Recall	F1-score	Support
0 (No diabetes)	0.86	0.73	0.79	100
1 (Diabetes)	0.61	0.78	0.68	54
Accuracy			0.75	154
Macro avg	0.73	0.75	0.74	154
Weighted avg	0.77	0.75	0.75	154

Confusion matrix:



The confusion matrix shows 73 true negatives, 27 false positives, 12 false negatives, and 42 true positives.

Model coefficients:

Feature	Coefficient
Glucose	0.904388
Insulin	0.849074
BMI	0.473949
Pregnancies	0.318641
SkinThickness	0.277745
DiabetesPedigreeFunction	0.227935
Age	0.164487
BloodPressure	0.023701

Glucose and Insulin carry the strongest positive coefficients, followed by BMI and Pregnancies, meaning higher values of these features are associated with a higher predicted diabetes risk within this trained model. BloodPressure has a coefficient close to zero.

8. Key Findings

- Glucose is the strongest driver of diabetes risk - it has the highest correlation with Outcome (Chart 4) and the largest model coefficient, followed closely by Insulin.
- BMI and Pregnancies also contribute positively, while BloodPressure has almost no influence in this model.
- The classes are imbalanced (about 65% vs 35%), so the model was trained with balanced class weights.
- The model reaches an accuracy of 74.7%, a recall of 77.8% and a ROC-AUC of 0.827 on the test set - a reasonable baseline for a screening-oriented classification task.
- Of the 54 actual diabetes cases in the test set, 42 were correctly identified and 12 were missed.

Limitations:

- The results are intended for learning and model-analysis purposes and should not be treated as a medical diagnosis.
- Precision is 60.9%: 27 non-diabetic cases in the test set were flagged as diabetic (false positives).
- Metrics come from a single fixed test split of 154 rows, so they may change with a different split.
- The coefficients show association within this trained model; they do not prove cause and effect.

9. Conclusion

This project applied a full logistic regression workflow - data inspection, exploratory analysis, invalid-zero treatment, median imputation, feature scaling, model training, evaluation, and coefficient interpretation - to the Pima Indians Diabetes dataset. The balanced Logistic Regression model achieved an accuracy of about 74.7%, precision of 60.9%, recall of 77.8%, F1-score of 68.3%, and ROC-AUC of 0.827. The recall shows that the model identified a relatively large proportion of the positive diabetes cases, which is useful for a screening-oriented task. Glucose and Insulin were the most influential features, followed by BMI and Pregnancies.